



FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3213-02

BUSINESS INTELLIGENCE

PROJECT PHASE 2

Title:

Retail Strategy: The Impact of GDP and Urbanization on European Sales

Lecturer: Dr Noorfa Haszlinna binti Mustaffa

Group: lilPotato

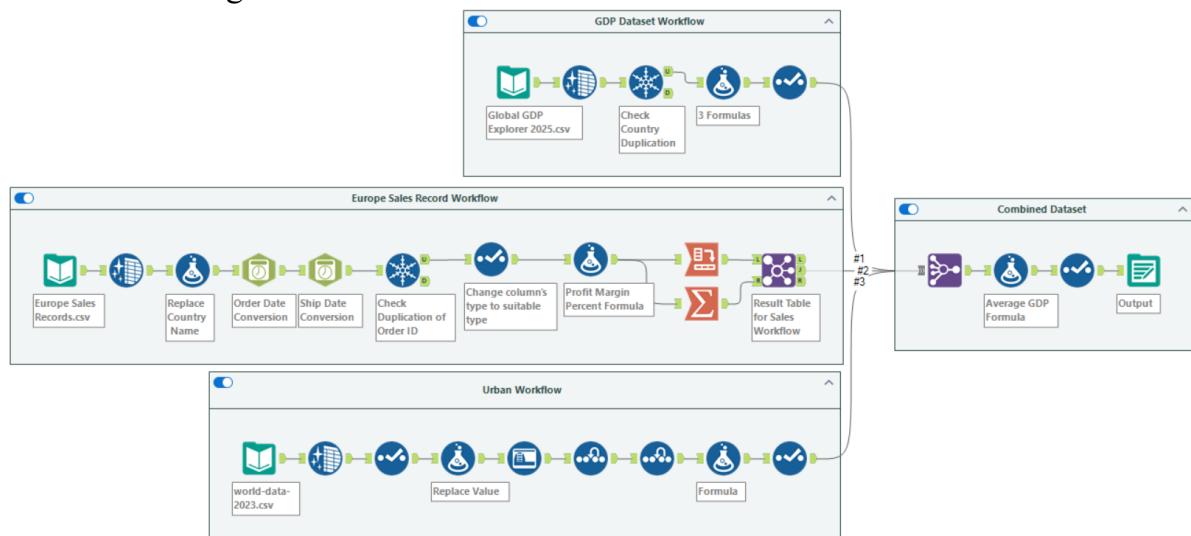
NO	NAME	MATRIC NO
1.	CHAU YING JIA	A23CS0213
2.	LAU YEE WEN	A23CS0099
3.	POH LOK YEE	A23CS0262

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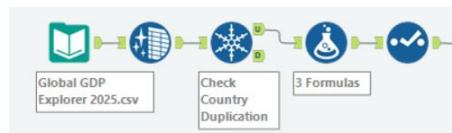
1.0 Workflow with Explanation of Tool Used

1.1 Overall Design



The complete Business Intelligence workflow is built within Alteryx Designer as a structured ETL (Extract, Transform, Load) pipeline. As shown in figure above, the workflow is organized into four sections, each enclosed within a Tool Container. The first container handles the GDP Dataset, the second processes the Europe Sales Record, and the third manages the Urbanisation Dataset. Each container independently cleans, standardizes, and prepares its respective dataset before passing the output into the fourth container, the Combined Dataset stage where all three cleaned streams are merged into a single master dataset, enriched with derived KPIs, and exported as an Excel file for Power BI visualization in Phase 3.

1.2 GDP Dataset Workflow

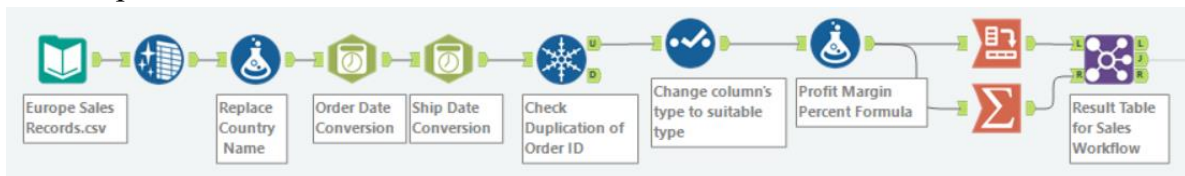


This section focuses on the Europe Sales Record Dataset. This workflow aims to clean and standardize the raw GDP data, resolve formatting issues and text encoding errors, so it can be joined with the other two datasets using Country as the common key. Table below explains the tools used for execution from left to right of the workflow.

No	Alteryx Tool	Explanation
1	Input data	Loads and reads the raw Global GDP Explorer 2025 CSV file into the data stream
2	Data Cleansing	Used to handles null values across all fields and trims hidden blank characters that may cause country name mismatches during join. GDP (nominal,2023), Population 2023 and GDP per capita columns were selected to remove the letters, punctuation, leading and all whitespaces.
3	Unique	Remove duplicate country entries since country field has been designed to serve as the unique identifier. Each country should appear only once.
4	Formula	Used to resolve three data quality issues simultaneously. First, the GDP Growth column contains a corrupted character â caused by a UTF-8 encoding error when the minus sign "-" was read incorrectly. The formula <code>Replace(Replace([GDP</code>

		Growth], "â", "-"), "%", "") first replaces the corrupted character with a proper minus sign, then removes the % symbol so the field can later be cast to a numeric type. Second, the Replace([Share of World GDP], "%", "") formula is used in Share of World GDP column to remove % symbol to allow numeric conversion later. Third, the Country column is standardized by replacing & with “And” using Replace([Country], "&", " And ") to ensure country names align consistently with the Europe Sales and Urbanisation datasets.
5	Select	Used to rename columns, change correct data types, and drop unnecessary fields. GDP (nominal, 2023) is renamed to GDP (\$) and cast to Double. GDP per capita is renamed to GDP per capita (\$) and cast to Double. Population 2023 is cast to Int64. GDP Growth is renamed to GDP Growth (%). Fields such as Field_1, GDP (abbrev.), and Share of World GDP are deselected and removed as they are not needed for analysis.

1.3 Europe Sales Record



This section specifically focuses on the Europe Sales Record Dataset. The primary goal of this workflow is to take raw, messy sales record data and turn it into a clean, highly structured format ready for integration with other two dataset. Table below explains the tool used for execution from the left to right of workflow.

No	Alteryx Tool	Explanation
1	Input data	Loads and reads the raw Europe Sales Record CSV file into the data stream
2	Data Cleansing	Used to remove unwanted characters like whitespaces and hidden blanks.
3	Formula (first)	Used to clean up text anomalies in the Country column to make sure all the countries names are uniform and standardize with GDP Dataset and Urbanisation Dataset.
4	DateTime	Used to turn plain text strings of date into actual date formats. DateTime tool is used here to convert both Order Date and Ship Date into a standard date format for easier sales trends tracking over time.
5	Unique	Used to configure a unique value, duplicate rows will be routed to separate output. Order ID column was ticked as it should be unique to identify each transaction and prevent double counting.
6	Select	Used to change data types, rename columns or drop unneeded fields. For example, OrderID are set to integer and TotalRevenue are set to double. Incorrect data types might cause errors and misleading aggregation.
7	Formula (second)	Used to create a new column named Profit_Margin_Percent and using the formula: IF [Total Revenue] == 0 THEN 0 ELSE ([Total Profit] / [Total Revenue]) * 100 ENDIF. Profit margin percentage is a business metric that is needed for performance analysis later.
8	Cross Tab	It pivots data from a long vertical list into a horizontal, summarized table layout. Used to count exactly how many orders came from Online versus Offline channels for each country.
9	Summarize	It groups data together to calculate totals, count, averages and other calculations. In this case, it is used to calculate the Total Profit and Total Revenue of each Country. The columns name can be renamed to Sum_Total Profit and Sum Total Revenue respectively. Lastly, the average of profit

		margin percentage is calculated and named it as Avg_Profit_Margin_Percent. This tool summarize thousands of rows into a clean summary.
10	Join	Join merges two separate data stream using a matching key. In this case, it takes the summarized table of channels preferability and the aggregated country stream and join them together using Country as the common field to link.

1.4 Urban Workflow

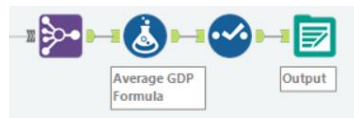


This section focuses on the Global Country Information Dataset 2023 (world-data-2023.csv). The workflow cleans and standardises country-level socio-economic data, resolves numeric formatting issues, handles missing values, and derives the Urban Population Percentage metric before joining with the other two datasets using Country as the common key. Table below explains the tools used for execution from left to right of the workflow.

No	Alteryx Tool	Explanation
1	Input data	Loads the raw world-data-2023.csv file into the workflow. All columns are read as V_String at this stage, including numeric fields such as Population, Urban_population, GDP, and Minimum wage.
2	Data Cleansing	Applied exclusively to the Country field, which serves as the join key across all three datasets. The tool is configured to remove leading and trailing whitespace, replace null rows with blanks or 0, and apply Title Case formatting. For example, a country name stored as " united kingdom " is standardised to "United Kingdom". This ensures that country names align correctly with those in the GDP and Europe Sales Record datasets during the Join Multiple step.
3	Select (first)	Retains only 10 relevant fields (Country, GDP, Population, Urban_population, Minimum wage, CPI, Life expectancy, Density, Land Area, Fertility Rate) and drops over 20 irrelevant columns such as Calling Code, Official Language, and Gasoline Price.
4	Formula (first)	Strips currency symbols and comma separators from GDP, Minimum wage, Population, and Urban_population using nested Replace() functions before converting them to numeric values with ToNumber(). For example, "\$1,234,567" becomes 1234567.
5	Auto Field	Automatically assigns the most memory-efficient data type to each field. For example, Population and Urban_population are reassigned from V_String to Int32 after the Formula tool conversion.
6	Imputation (First)	Replaces null values in six numeric fields using mean imputation: Density (P/Km2), Land Area(Km2), CPI, Fertility Rate, Life expectancy, and GDP_Clean. Null values arise for small territories or nations where complete statistical data is unavailable. Rather than dropping these rows and losing the country from the join, each null is replaced with the calculated mean of all non-null values in that specific field. This preserves as many country records as possible for the final integrated dataset.
7	Imputation (Second)	A second imputation pass is applied separately to Urban_population and Minimum wage_clean. Unlike the first pass which targets null values, this pass is configured to replace zero (0) values. After the Formula tool's string-to-number conversion in Step 4, records where the original string was empty

		or contained only formatting characters may have been converted to 0 rather than null. Replacing these zeroes with the mean prevents them from distorting aggregations and KPI calculations such as Urban Population Percentage downstream.
8	Formula (Second)	Creates a derived feature, Urban Population Percentage, using the formula: IF [Population] = 0 THEN NULL() ELSE ([Urban_population] / [Population]) * 100 ENDIF. This metric is used later to calculate the primary KPI, Profit per Urban Resident.
9	Select (Second)	Drops the original raw string fields and renames the clean numeric versions (GDP_Clean → GDP, Minimum wage_clean → Minimum wage, Urban_Population_Pct → Urban Population Percentage) before passing the stream to the Join Multiple tool.

1.5 Combined Dataset Workflow



This section covers the final integration stage where all three cleaned datasets are merged into one unified master dataset. Additional KPI derivation and final column selection are also performed here before the data is exported. Table below explains the tools used for execution from left to right of the workflow.

No	Alteryx Tool	Explanation
1	Join Multiple	Merges all three cleaned data streams using Country columns as the common joining key across all three inputs. Unlike a standard Join tool which only handles two streams, the Join Multiple tool allows all three datasets to be combined in a single step. The result is one unified row per country containing sales, GDP, and urbanisation fields together.
2	Formula	Used to create five new calculated fields that provide deeper business insight beyond the raw data. First, Average GDP is calculated using Average([GDP (\$)], [GDP]) to combine the GDP values from both the GDP dataset and Urbanisation dataset into a single representative figure per country. Second, Average Population uses Average([Population 2023], [Population]) for the same reason as both source datasets contain a population figure and averaging them produces a more balanced value. Third, the primary KPI of this project, Profit per Urban Resident is calculated using [Sum_Total Profit] / [Urban Population Percentage] to measure how efficiently each country generates profit relative to its urban population. Fourth, Channel Preference classifies each country as "Online", "Offline", or "Balanced" based on whether Count_Online is greater than, less than, or equal to Count_Offline. Lastly, GDP Tier categorises each country into "High (>\$50k)", "Medium (\$25k–\$50k)", or "Low (<\$25k)" based on GDP per capita (\$) to enable economic segmentation in the Power BI dashboard.
3	Select	Used to perform the final dimensional reduction and clean-up of the master dataset. Redundant source columns such as the original GDP (\$), Population 2023, GDP, and Population from individual datasets are deselected and removed since they have been replaced by the averaged GDP and Population fields. Some columns are also renamed for clarity, for example, Count_Offline becomes Offline Preferability, Count_Online becomes Online Preferability, and Avg_Profit_Margin_Percent becomes Profit Margin (%).

4	Output Data	Exports the final cleaned and integrated master dataset as an .xlsx file (Excel format). This file is the direct data source loaded into Microsoft Power BI for dashboard creation and visualization in Phase 3.
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2.0 Challenges Faced

Challenge 1: Preventing Division-by-Zero Calculation Failures

When creating the Profit_Margin_Percent metric for the Europe Sales Record Dataset, the math requires dividing total profit by total revenue. However, if any specific row happens to have a revenue of exactly zero like a cancelled order. The result will throw an error or leave a blank space. To fix this, a conditional statement inside the Formula tool has to be written that checks if the revenue is zero first. If it is, it just sets the margin to 0%, otherwise, it safely runs the calculation.

Challenge 2: Managing Irrelevant Columns Causing Schema Confusion

The Global Country Information Dataset 2023 contains over 30 columns covering a wide range of indicators such as Calling Code, Official Language, Gasoline Price, Forested Area (%), and Co2-Emissions, most of which are not relevant to the business problem. Allowing all these columns to pass through into the Join Multiple tool would bloat the master dataset schema, making it harder to identify meaningful fields in Power BI and risking column name conflicts with fields from the other two datasets. To resolve this, the Select tool was applied twice throughout the Urban Workflow. The first Select immediately after the Data Cleansing step reduced the dataset from over 30 columns to only 10 relevant fields. The second Select at the end of the pipeline performed a final clean-up by removing the original raw string versions of GDP and Minimum wage after their cleaned numeric equivalents were created. This two-stage column reduction kept the schema clean and ensured only meaningful fields entered the final integrated dataset.

Challenge 3: Resolution of Duplicate Population and GDP Fields After Joining

After combining all three datasets with Join multiple tool, the merged table included two separate columns for both population and GDP figures from GDP datasets and the Urbanisation dataset. Having two columns representing the same real-world metric for the same country caused inconsistency and would cause confusion during Power BI visualization. Dropping one at random was also not optimal as both sources could give different values due to different data collection periods and methodology. The solution was applying the Average () function to calculate the average GDP and average population as the mean of both source values per country, producing a single balanced representative figure. Both original duplicate columns were then deselected in the Select tool so that it kept only the averaged versions in the master output.